



DC – AI – Routing – Switching – Security – Wireless – Automation

VPLS (LDP/BGP) to EVPN migration

An HPE Juniper Networking Techpost

Ramdas Machat

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HPE



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1 VPLS to EVPN-VPLS Seamless Migration on MX Routers

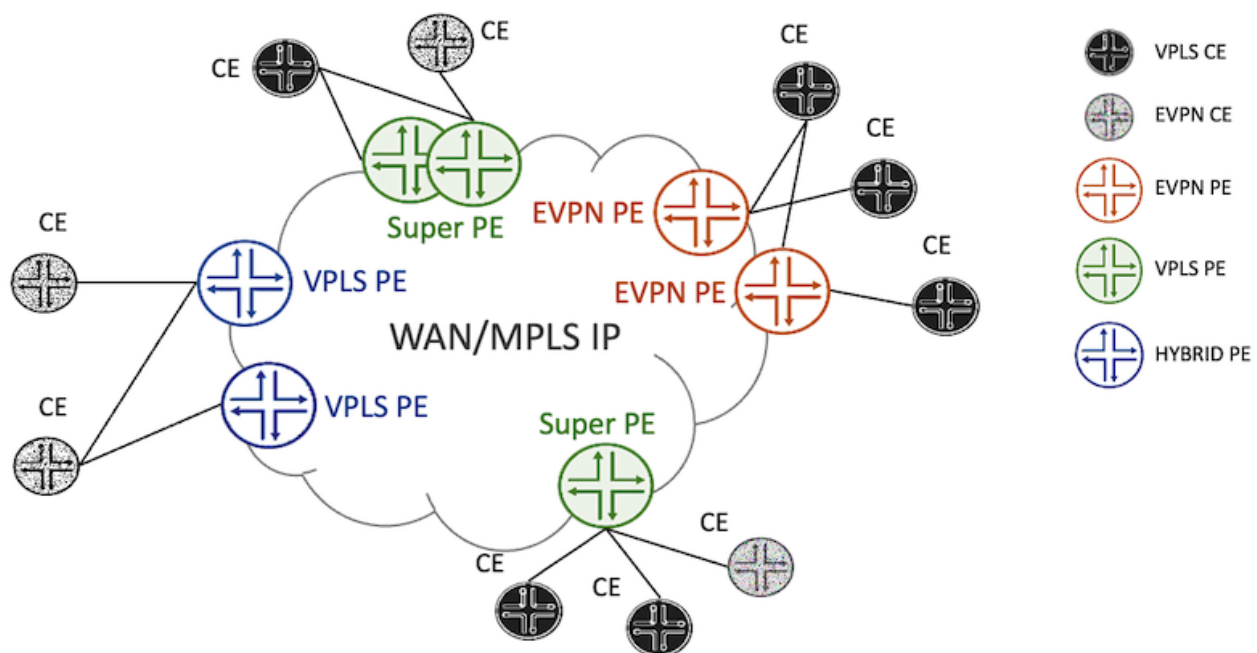
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Techniques, configurations and best practices for migrating from legacy business services to EVPN on MX Routers.

1.1 Introduction

Service providers started adopting the EVPN technology as a replacement for their L2 technology requirements like VPLS, L2VPN. There is a lot of benefits of EVPN as a technology which is briefly covered in the below section. At the same time service providers also look for the migration of the legacy L2 technologies (which has with a lot of limitations) to the EVPN technology.

This solution provides a lot of flexibility to the service providers in doing a staged migration and the best part is the coexistence of the legacy technologies and EVPN at the same time as described in Figure 1. So, PEs in different segments can run VPLS and EVPN for the same VPN routing instance. Also, the migration supports all the different scenarios like single-homed / multi-homed PEs.



The [TechLibrary article linked here](#) describes the migration from BGP VPLS to EVPN.

EVPN is a next-generation technology helping the service providers to overcome the limitations of the legacy L2 business services. It's a more efficient technology with better control plane, the data plane learning and programming capabilities.

EVPN over MPLS is considered the main enabler of the Carrier Ethernet services in the Metro Area Networks (MAN) and Mobile BackHaul (MBH) networks by many service providers and is the integral attribute of the Cloud Metro Architecture. It covers all possible overlay service topology: point-to-point, multipoint-to-multipoint and point-to-multipoint, seamlessly. The table below shows the mapping of Layer 2 service definitions with different industry stands and technology including EVPN.

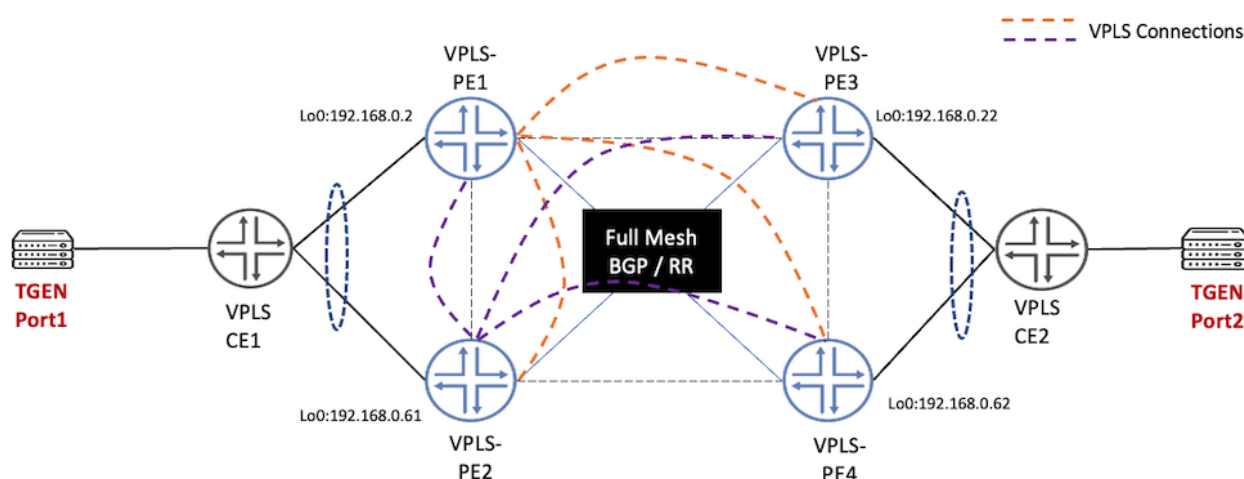
Standard	Point-to-Point	Multipoint-to-Multipoint	Point-to-Multipoint
MEF	E-LINE	E-LAN	E-TREE
IETF	VPWS (Virtual Private Wire Service)	VPLS (Virtual Private LAN Service)	Achieved using Flooding mechanism
Martini (LDP)	L2Circuit	VPLS	H-VPLS
BGP	L2VPN/FEC-129	VPLS/FEC-129	H-VPLS/FEC-129
EVPN	EVPN-EVPS	EVPN	EVPN E-Tree

1.1.1 Seamless Migration: Best practices

Let's pick a VPLS PE router with multiple instances. The idea of "seamless migration" represents the capability of migrating some instances to EVPN technology while maintaining the VPLS operation on the other instances. The instance migrated to EVPN can also operate with a VPLS instance till the other end VPLS PE is also migrated to EVPN.

Here are some best practices for migration.

1. In a scaled environment, pick the VPLS PE instances with lower MAC count and least traffic. Do the migration (procedure in details given below) and wait for all the MAC addresses to be learnt and traffic is forwarded successfully.
2. When all the instances are migrated, the PE is completely migrated to EVPN: it is time to clean-up old configuration. For the BGP-VPLS clean-up, a BGP flap is expected because of signalling and negotiation of BGP capabilities.
3. Migrate all multihomed PEs to EVPN and do NOT leave some as VPLS-only and migrate others (as VPLS and EVPN use different algorithms to decide Designated forwarder). Multihomed PEs must be in sync regarding the forwarding decision of BUM traffic received from the core towards access.



	CE1	PE1	PE2	PE3	PE4	CE2
Model	qfx5110-48s-4c	MX480	MX10008	MX240	MX960	qfx5200-32c-32q
JUNOS Version	20.2R2.11	22.1R2	22.1R2	22.1R2	22.1R2	21.3R1.9

1.2 BGP-Signaled VPLS to EVPN Migration

1.2.1 BGP-VPLS Configuration and Operational status

All four PEs in the above topology are BGP-VPLS PE and have the VPLS instance setup and then relevant configurations are made in each one of these PEs. These PEs discover each other in the network and establishes the VPLS sessions. The configuration and operational status on two of the PE1 and PE4 are shown below, the rest is removed for sake of brevity.

PE1 Configuration:

```

1 regress@rtme-mx-02# show protocols bgp
2 group iBGP {
3     type internal;
4     local-address 192.168.0.2;
5     family inet {
6         unicast;
7     }
8     family inet-vpn {
9         unicast;
10    }
11    family l2vpn {
12        signaling;
13    }
14    neighbor 192.168.0.22;
15    neighbor 192.168.0.62;
16    neighbor 192.168.0.61;
17 }
18
19 regress@rtme-mx-02# show routing-instances vpls-bgp
20 instance-type vpls;
21 protocols {
22     vpls {
23         site site1 {
24             site-identifier 2;
25         }
26         no-tunnel-services;
27     }

```

PE4 Configuration:

```

1 regress@rtme-mx-62-1# show protocols bgp
2 group iBGP {
3     type internal;
4     local-address 192.168.0.62;
5     family inet {
6         unicast;
7     }
8     family inet-vpn {
9         unicast;
10    }
11    family l2vpn {
12        signaling;
13    }
14    neighbor 192.168.0.2;
15    neighbor 192.168.0.61;
16    neighbor 192.168.0.22;
17 }

```

```

18
19
20 regress@rtme-mx-62-1# show routing-instances vpls-bgp
21 instance-type vpls;
22 protocols {
23     vpls {
24         site site1 {
25             site-identifier 62;
26         }
27         no-tunnel-services;
28     }
29 }
30 interface et-5/0/2.1;
31 route-distinguisher 192.168.0.62:1;
32 vrf-target target:1:1;
33
34
35 regress@rtme-mx-62-1# show interfaces et-5/0/2
36 description acces-qfx15;
37 flexible-vlan-tagging;
38 encapsulation flexible-ethernet-services;
39 unit 1 {
40     encapsulation vlan-vpls;
41     vlan-id 1234;
42 }

```

PE1 operational status:

```

1 regress@rtme-mx-02# run show vpls connections
2 Layer-2 VPN connections:
3 Instance: vpls-bgp
4 Edge protection: Not-Primary
5 Local site: site1 (2)
6 connection-site Type St Time last up # Up trans
7 22 rmt Up Feb 13 05:43:01 2022 1
8 Remote PE: 192.168.0.22, Negotiated control-word: No
9 Incoming label: 78, Outgoing label: 70
10 Local interface: lsi.1048576, Status: Up, Encapsulation: VPLS
11 Description: Intf - vpls vpls-bgp local site 2 remote site 22
12 Flow Label Transmit: No, Flow Label Receive: No
13 61 rmt Up Feb 13 05:43:01 2022 1
14 Remote PE: 192.168.0.61, Negotiated control-word: No
15 Incoming label: 85, Outgoing label: 55
16 Local interface: lsi.1048577, Status: Up, Encapsulation: VPLS
17 Description: Intf - vpls vpls-bgp local site 2 remote site 61
18 Flow Label Transmit: No, Flow Label Receive: No
19 62 rmt Up Feb 13 05:43:01 2022 1
20 Remote PE: 192.168.0.62, Negotiated control-word: No
21 Incoming label: 86, Outgoing label: 48
22 Local interface: lsi.1048578, Status: Up, Encapsulation: VPLS
23 Description: Intf - vpls vpls-bgp local site 2 remote site 62
24 Flow Label Transmit: No, Flow Label Receive: No
25
26
27 regress@rtme-mx-02# run show bgp summary
28 Threading mode: BGP I/O
29 Default eBGP mode: advertise - accept, receive - accept
30 Groups: 2 Peers: 3 Down peers: 0
31 Table Tot Paths Act Paths Suppressed History Damp State Pending
32 inet.0
33 0 0 0 0 0 0
34 bgp.l3vpn.0
35 0 0 0 0 0 0
36 lsdist.0
37 0 0 0 0 0 0
38 bgp.l2vpn.0
39 9 9 0 0 0 0
40 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
41 192.168.0.22 65000 327 337 0 0 2:22:59 Establ
42 inet.0: 0/0/0/0
43 bgp.l3vpn.0: 0/0/0/0
44 bgp.l2vpn.0: 3/3/3/0
45 192.168.0.61 65000 3762 334 0 0 2:22:59 Establ
46 inet.0: 0/0/0/0
47 bgp.l3vpn.0: 0/0/0/0
48 bgp.l2vpn.0: 3/3/3/0
49 192.168.0.62 65000 336 336 0 0 2:22:59 Establ
50 inet.0: 0/0/0/0
51 bgp.l3vpn.0: 0/0/0/0
52 bgp.l2vpn.0: 3/3/3/0

```

PE4 operational status:

```

1 regress@rtme-mx-62-1# run show vpls connections
2 Layer-2 VPN connections:
3 Instance: vpls-bgp
4 Edge protection: Not-Primary
5 Local site: site1 (62)
6 connection-site Type St Time last up # Up trans
7 2 rmt Up Feb 13 05:43:01 2022 1
8 Remote PE: 192.168.0.2, Negotiated control-word: No
9 Incoming label: 48, Outgoing label: 86
10 Local interface: lsi.1048578, Status: Up, Encapsulation: VPLS
11 Description: Intf - vpls vpls-bgp local site 62 remote site 2
12 Flow Label Transmit: No, Flow Label Receive: No
13 22 rmt Up Feb 13 05:36:29 2022 1
14 Remote PE: 192.168.0.22, Negotiated control-word: No
15 Incoming label: 44, Outgoing label: 66
16 Local interface: lsi.1048576, Status: Up, Encapsulation: VPLS
17 Description: Intf - vpls vpls-bgp local site 62 remote site 22
18 Flow Label Transmit: No, Flow Label Receive: No
19 61 rmt Up Feb 13 05:44:30 2022 1
20 Remote PE: 192.168.0.61, Negotiated control-word: No
21 Incoming label: 35, Outgoing label: 43
22 Local interface: lsi.1048579, Status: Up, Encapsulation: VPLS
23 Description: Intf - vpls vpls-bgp local site 62 remote site 61
24 Flow Label Transmit: No, Flow Label Receive: No
25
26 [edit]
27 regress@rtme-mx-62-1# run show bgp summary
28 Threading mode: BGP I/O
29 Default eBGP mode: advertise - accept, receive - accept
30 Groups: 4 Peers: 7 Down peers: 2
31 Table Tot Paths Act Paths Suppressed History Damp State Pending
32 inet.0
33 0 0 0 0 0 0
34 bgp.l3vpn.0
35 0 0 0 0 0 0
36 bgp.l2vpn.0
37 9 9 0 0 0 0
38 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
39 192.168.0.2 65000 339 337 0 2 2:24:14 Establ
40 inet.0: 0/0/0/0
41 bgp.l3vpn.0: 0/0/0/0
42 bgp.l2vpn.0: 3/3/3/0
43 192.168.0.22 65000 662 662 0 0 4:52:52 Establ
44 inet.0: 0/0/0/0
45 bgp.l3vpn.0: 0/0/0/0
46 bgp.l2vpn.0: 3/3/3/0
47 192.168.0.61 65000 32 30 0 0 10:19 Establ
48 inet.0: 0/0/0/0
49 bgp.l3vpn.0: 0/0/0/0
50 bgp.l2vpn.0: 3/3/3/0

```

1.2.2 PE1 Instance Migration from BGP-VPLS to EVPN

All the four PEs in the topology in Figure 2 are BGP-VPLS PEs. We will perform the migration for one of them, from VPLS to EVPN. Once completed, we will have VPLS PE on one side and EVPN PE on the other.

Migration of PE1: We add the protocol evpn details under the routing-instance and we add the family evpn signalling:

```

1 regress@rtme-mx-02# show routing-instances vpls-bgp
2 instance-type evpn;
3 protocols {
4   evpn { <<< Add this part to the routing instance and keep the VPLS config
5     interface et-0/1/2.1;
6   }
7   vpls {
8     site site1 {
9       site-identifier 2;
10    }

```

```

11     no-tunnel-services;
12 }
13 }
14 vlan-id 1234;
15 interface et-0/1/2.1;
16 route-distinguisher 192.168.0.2:1;
17 vrf-target target:1:1;
18
19 regress@rtme-mx-02# show protocols bgp group iBGP
20 type internal;
21 local-address 192.168.0.2;
22 family inet {
23     unicast;
24 }
25 family inet-vpn {
26     unicast;
27 }
28 family l2vpn {
29     signaling;
30 }
31 family evpn { <<< Add this to signal the BGP with evpn family to all the neighbours.
32     <<< You can also specifically add this under the desired neighbour for signalling.
33     signaling;
34 }
35 neighbor 192.168.0.22;
36 neighbor 192.168.0.62;
37 neighbor 192.168.0.61;

```

PE1 still maintains the VPLS status with the other PE, however, the forwarding is now happening using the EVPN technology. We can see the new EVPN tables are created, and the EVPN routes are exchanged between the PEs.

```

1 regress@rtme-mx-02# run show vpls connections
2 Layer-2 VPN connections:
3
4 Instance: vpls-bgp
5 Edge protection: Not-Primary
6 Local site: sitel (2)
7 connection-site Type St Time last up # Up trans
8 22 rmt Up Feb 13 07:33:32 2022 1
9 Remote PE: 192.168.0.22, Negotiated control-word: No
10 Incoming label: 117, Outgoing label: 87
11 Local interface: lsi.1049093, Status: Up, Encapsulation: VPLS
12 Description: Intf - vpls vpls-bgp local site 2 remote site 22
13 Flow Label Transmit: No, Flow Label Receive: No
14 61 rmt Up Feb 13 07:33:32 2022 1
15 Remote PE: 192.168.0.61, Negotiated control-word: No
16 Incoming label: 124, Outgoing label: 79
17 Local interface: lsi.1049094, Status: Up, Encapsulation: VPLS
18 Description: Intf - vpls vpls-bgp local site 2 remote site 61
19 Flow Label Transmit: No, Flow Label Receive: No
20 62 rmt Up Feb 13 07:33:32 2022 1
21 Remote PE: 192.168.0.62, Negotiated control-word: No
22 Incoming label: 125, Outgoing label: 67
23 Local interface: lsi.1049095, Status: Up, Encapsulation: VPLS
24 Description: Intf - vpls vpls-bgp local site 2 remote site 62
25 Flow Label Transmit: No, Flow Label Receive: No
26
27
28 regress@rtme-mx-02# run show bgp summary
29 Threading mode: BGP I/O
30 Default eBGP mode: advertise - accept, receive - accept
31 Groups: 2 Peers: 3 Down peers: 0
32 Table Tot Paths Act Paths Suppressed History Damp State Pending
33 inet.0
34 0 0 0 0 0 0
35 bgp.l3vpn.0
36 0 0 0 0 0 0
37 bgp.evpn.0
38 0 0 0 0 0 0
39 lsdist.0
40 0 0 0 0 0 0
41 bgp.l2vpn.0
42 9 9 0 0 0 0
43 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
44 192.168.0.22 65000 176 178 0 0 1:14:49 Establ
45 inet.0: 0/0/0/0
46 bgp.l3vpn.0: 0/0/0/0

```

```

47  bgp.l2vpn.0: 3/3/3/0
48  vpls-bgp.l2vpn.0: 3/3/3/0
49  192.168.0.61 65000 176 177 0 0 1:14:49 Establ
50  inet.0: 0/0/0/0
51  bgp.l3vpn.0: 0/0/0/0
52  bgp.l2vpn.0: 3/3/3/0
53  vpls-bgp.l2vpn.0: 3/3/3/0
54  192.168.0.62 65000 112 109 0 2 44:49 Establ
55  inet.0: 0/0/0/0
56  bgp.l3vpn.0: 0/0/0/0
57  bgp.l2vpn.0: 3/3/3/0
58  vpls-bgp.l2vpn.0: 3/3/3/0
59
60
61  regress@rtme-mx-02# run show evpn instance
62                               Intfs IRB intfs MH MAC addresses
63  Instance Total Up Total Up Nbrs ESIs Local Remote
64  __default_evpn__ 0
65  vpls-bgp 2 2 0 0 0 0 0 0
66
67  [edit]
68  regress@rtme-mx-02# run show evpn instance extensive
69  Instance: __default_evpn__
70  Route Distinguisher: 192.168.0.2:0
71  Number of bridge domains: 0
72  Number of neighbors: 0
73
74  Instance: vpls-bgp
75  Route Distinguisher: 192.168.0.2:1
76  VLAN ID: 1234
77  Per-instance MAC route label: 102
78  Duplicate MAC detection threshold: 5
79  Duplicate MAC detection window: 180
80  MAC database status Local Remote
81  MAC advertisements: 0 0
82  MAC+IP advertisements: 0 0
83  Default gateway MAC advertisements: 0 0
84  Number of local interfaces: 2 (2 up)
85  Interface name ESI Mode Status AC-Role
86  .local..8 00:00:00:00:00:00:00:00:00:00 single-homed Up Root
87  et-0/1/2.1 00:00:00:00:00:00:00:00:00:00 single-homed Up Root
88  Number of IRB interfaces: 0 (0 up)
89  Number of protect interfaces: 0
90  Number of bridge domains: 1
91  VLAN Domain-ID Intfs/up IRB-intf Mode MAC-sync IM-label MAC-label v4-SG-sync IM-core-NH v6-SG
    -sync IM-core-NH Trans-ID
92  1234 1 1 Extended Enabled 128 Disabled Disabled
93  Number of neighbors: 0
94  Number of ethernet segments: 0
95  SMET Forwarding: Disabled
96  Migration: BGP-VPLS
97
98
99  regress@rtme-mx-02# run show route table bgp.evpn.0
100
101  bgp.evpn.0: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)
102  + = Active Route, - = Last Active, * = Both
103
104  3:192.168.0.2:1::1234::192.168.0.2/248 IM
105      *[EVPN/170] 00:10:27
106      Indirect

```

PE2, PE3, and PE4 continue to be VPLS PEs, so no change in configuration.

Below is the operational status of PE4, just as a reference that this PE still has the business service up using VPLS on its end while PE1 is now EVPN.

```

1  regress@rtme-mx-62-1# run show vpls connections
2  Layer-2 VPN connections:
3
4  Instance: vpls-bgp
5  Edge protection: Not-Primary
6  Local site: site1 (62)
7  connection-site Type St Time last up # Up trans
8  2 rmt Up Feb 13 07:33:32 2022 1
9  Remote PE: 192.168.0.2, Negotiated control-word: No
10 Incoming label: 67, Outgoing label: 125
11 Local interface: lsi.1048839, Status: Up, Encapsulation: VPLS

```

```

12      Description: Intf - vpls vpls-bgp local site 62 remote site 2
13      Flow Label Transmit: No, Flow Label Receive: No
14      22 rmt Up Feb 13 07:33:14 2022 1
15      Remote PE: 192.168.0.22, Negotiated control-word: No
16      Incoming label: 79, Outgoing label: 99
17      Local interface: lsi.1048837, Status: Up, Encapsulation: VPLS
18      Description: Intf - vpls vpls-bgp local site 62 remote site 22
19      Flow Label Transmit: No, Flow Label Receive: No
20      61 rmt Up Feb 13 07:33:14 2022 1
21      Remote PE: 192.168.0.61, Negotiated control-word: No
22      Incoming label: 62, Outgoing label: 75
23      Local interface: lsi.1048838, Status: Up, Encapsulation: VPLS
24      Description: Intf - vpls vpls-bgp local site 62 remote site 61
25      Flow Label Transmit: No, Flow Label Receive: No

```

1.2.3 PE3: BGP-VPLS Instance to EVPN Instance Migration

The below section shows the migration of PE3 (VPLS PE) to EVPN PE and checks how it learns the routes from PE1 which is already moved to EVPN technology.

```

1 regress@rtme-mx-22# show routing-instances vpls-bgp
2 instance-type evpn;
3 protocols {
4     evpn {
5         interface et-0/0/2.1;
6     }
7     vpls {
8         site site1 {
9             site-identifier 22;
10        }
11        no-tunnel-services;
12    }
13 }
14 vlan-id 1234;
15 interface et-0/0/2.1;
16 route-distinguisher 192.168.0.22:1;
17 vrf-target target:1:1;
18
19 [edit]
20 regress@rtme-mx-22# show protocols bgp group iBGP
21 type internal;
22 local-address 192.168.0.22;
23 family inet {
24     unicast;
25 }
26 family inet-vpn {
27     unicast;
28 }
29 family l2vpn {
30     signaling;
31 }
32 family evpn {
33     signaling;
34 }
35 neighbor 192.168.0.2;
36 neighbor 192.168.0.61;
37 neighbor 192.168.0.62;

```

PE3 has established an EVPN PE neighborhood with PE1 and has learnt the EVPN routes. At the same time, the other two PEs which are still VPLS PEs maintain the VPLS control plane and forwarding state.

```

1 regress@rtme-mx-22# run show vpls connections
2 Layer-2 VPN connections:
3
4 Instance: vpls-bgp
5 Edge protection: Not-Primary
6 Local site: site1 (22)
7 connection-site Type St Time last up # Up trans
8 2 rmt Up Feb 13 07:51:04 2022 1
9 Remote PE: 192.168.0.2, Negotiated control-word: No
10 Incoming label: 108, Outgoing label: 117
11 Local interface: lsi.1048593, Status: Up, Encapsulation: VPLS
12 Description: Intf - vpls vpls-bgp local site 22 remote site 2
13 Flow Label Transmit: No, Flow Label Receive: No

```

```

14 61 rmt Up Feb 13 07:51:04 2022 1
15 Remote PE: 192.168.0.61, Negotiated control-word: No
16 Incoming label: 98, Outgoing label: 91
17 Local interface: lsi.1048594, Status: Up, Encapsulation: VPLS
18 Description: Intf - vpls vpls-bgp local site 22 remote site 61
19 Flow Label Transmit: No, Flow Label Receive: No
20 62 rmt Up Feb 13 07:51:04 2022 1
21 Remote PE: 192.168.0.62, Negotiated control-word: No
22 Incoming label: 99, Outgoing label: 79
23 Local interface: lsi.1048595, Status: Up, Encapsulation: VPLS
24 Description: Intf - vpls vpls-bgp local site 22 remote site 62
25 Flow Label Transmit: No, Flow Label Receive: No
26
27 [edit]
28 regress@rtme-mx-22# run show bgp summary
29 Threading mode: BGP I/O
30 Default eBGP mode: advertise - accept, receive - accept
31 Groups: 2 Peers: 3 Down peers: 0
32 Table Tot Paths Act Paths Suppressed History Damp State Pending
33 inet.0
34 0 0 0 0 0 0
35 bgp.l3vpn.0
36 0 0 0 0 0 0
37 bgp.evpn.0
38 1 1 0 0 0 0
39 lsdist.0
40 0 0 0 0 0 0
41 bgp.l2vpn.0
42 9 9 0 0 0 0
43 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
44 192.168.0.2 65000 15 17 0 0 4:01 Establ
45 inet.0: 0/0/0/0
46 bgp.l3vpn.0: 0/0/0/0
47 bgp.evpn.0: 1/1/1/0
48 __default_evpn__.evpn.0: 0/0/0/0
49 bgp.l2vpn.0: 3/3/3/0
50 vpls-bgp.evpn.0: 1/1/1/0
51 vpls-bgp.l2vpn.0: 3/3/3/0
52 192.168.0.61 65000 15 13 0 0 3:57 Establ
53 inet.0: 0/0/0/0
54 bgp.l3vpn.0: 0/0/0/0
55 bgp.l2vpn.0: 3/3/3/0
56 vpls-bgp.l2vpn.0: 3/3/3/0
57 192.168.0.62 65000 16 15 0 0 4:01 Establ
58 inet.0: 0/0/0/0
59 bgp.l3vpn.0: 0/0/0/0
60 bgp.l2vpn.0: 3/3/3/0
61 vpls-bgp.l2vpn.0: 3/3/3/0
62
63 regress@rtme-mx-22# run show evpn instance
64 Intfs IRB intfs MH MAC addresses
65 Instance Total Up Total Up Nbrs ESIs Local Remote
66 __default_evpn__ 0
67 vpls-bgp 2 1 0 0 1 0 0 0
68
69 [edit]
70 regress@rtme-mx-22# run show evpn instance extensive
71 Instance: __default_evpn__
72 Route Distinguisher: 192.168.0.22:0
73 Number of bridge domains: 0
74 Number of neighbors: 0
75
76 Instance: vpls-bgp
77 Route Distinguisher: 192.168.0.22:1
78 VLAN ID: 1234
79 Per-instance MAC route label: 102
80 Duplicate MAC detection threshold: 5
81 Duplicate MAC detection window: 180
82 MAC database status Local Remote
83 MAC advertisements: 0 0
84 MAC+IP advertisements: 0 0
85 Default gateway MAC advertisements: 0 0
86 Number of local interfaces: 2 (1 up)
87 Interface name ESI Mode Status AC-Role
88 .local..11 00:00:00:00:00:00:00:00:00 single-homed Up Root
89 et-0/0/2.1 00:00:00:00:00:00:00:00:00 single-homed Up Root
90 Number of IRB interfaces: 0 (0 up)
91 Number of protect interfaces: 0
92 Number of bridge domains: 1
93 VLAN Domain-ID Intfs/up IRB-intf Mode MAC-sync IM-label MAC-label v4-SG-sync IM-core-NH v6-SG
-synch IM-core-NH Trans-ID

```

```

94      1234 1 1 Extended Enabled 104 Disabled Disabled
95      Number of neighbors: 1
96      Address MAC MAC+IP AD IM ES Leaf-label Remote-DCI-Peer
97      192.168.0.2 0 0 0 1 0
98      Number of ethernet segments: 0
99      SMET Forwarding: Disabled
100     Migration: BGP-VPLS
101
102
103 regress@rtme-mx-22# run show route table bgp.evpn.0
104
105 bgp.evpn.0: 2 destinations, 2 routes (2 active, 0 holddown, 0 hidden)
106 + = Active Route, - = Last Active, * = Both
107
108 3:192.168.0.2:1::1234::192.168.0.2/248 IM
109      *[BGP/170] 00:05:26, localpref 100, from 192.168.0.2
110      AS path: I, validation-state: unverified
111      > to 11.0.0.30 via et-1/1/2.0, Push 43
112 3:192.168.0.22:1::1234::192.168.0.22/248 IM
113      *[EVPN/170] 00:05:27
114      Indirect

```

On PE1:

```

1 regress@rtme-mx-02# run show vpls connections
2 Layer-2 VPN connections:
3
4 Instance: vpls-bgp
5 Edge protection: Not-Primary
6 Local site: site1 (2)
7 connection-site Type St Time last up # Up trans
8 22 rmt Up Feb 13 07:51:04 2022 1
9 Remote PE: 192.168.0.22, Negotiated control-word: No
10 Incoming label: 117, Outgoing label: 108
11 Local interface: lsi.1049096, Status: Up, Encapsulation: VPLS
12 Description: Intf - vpls vpls-bgp local site 2 remote site 22
13 Flow Label Transmit: No, Flow Label Receive: No
14 61 rmt Up Feb 13 07:33:32 2022 1
15 Remote PE: 192.168.0.61, Negotiated control-word: No
16 Incoming label: 124, Outgoing label: 79
17 Local interface: lsi.1049094, Status: Up, Encapsulation: VPLS
18 Description: Intf - vpls vpls-bgp local site 2 remote site 61
19 Flow Label Transmit: No, Flow Label Receive: No
20 62 rmt Up Feb 13 07:33:32 2022 1
21 Remote PE: 192.168.0.62, Negotiated control-word: No
22 Incoming label: 125, Outgoing label: 67
23 Local interface: lsi.1049095, Status: Up, Encapsulation: VPLS
24 Description: Intf - vpls vpls-bgp local site 2 remote site 62
25 Flow Label Transmit: No, Flow Label Receive: No
26
27 [edit]
28 regress@rtme-mx-02# run show bgp summary
29 Threading mode: BGP I/O
30 Default eBGP mode: advertise - accept, receive - accept
31 Groups: 2 Peers: 3 Down peers: 0
32 Table Tot Paths Act Paths Suppressed History Damp State Pending
33 inet.0
34      0 0 0 0 0 0
35 bgp.l3vpn.0
36      0 0 0 0 0 0
37 bgp.evpn.0
38      1 1 0 0 0 0
39 lsdist.0
40      0 0 0 0 0 0
41 bgp.l2vpn.0
42      9 9 0 0 0 0
43 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
44 192.168.0.22 65000 24 20 0 1 6:09 Establ
45 inet.0: 0/0/0/0
46 bgp.l3vpn.0: 0/0/0/0
47 bgp.evpn.0: 1/1/1/0
48 evpn-vpws-vlan-aware.evpn.0: 0/0/0/0
49 __default_evpn__.evpn.0: 0/0/0/0
50 bgp.l2vpn.0: 3/3/3/0
51 vpls-bgp.l2vpn.0: 3/3/3/0
52 vpls-bgp.evpn.0: 1/1/1/0
53 192.168.0.61 65000 210 212 0 0 1:30:30 Establ
54 inet.0: 0/0/0/0
55 bgp.l3vpn.0: 0/0/0/0
56 bgp.l2vpn.0: 3/3/3/0

```

```

57 vpls-bgp.l2vpn.0: 3/3/3/0
58 192.168.0.62 65000 146 144 0 2 1:00:30 Establ
59 inet.0: 0/0/0/0
60 bgp.l3vpn.0: 0/0/0/0
61 bgp.l2vpn.0: 3/3/3/0
62 vpls-bgp.l2vpn.0: 3/3/3/0
63
64 regress@rtme-mx-02# run show evpn instance
65                               Intfs IRB intfs MH MAC addresses
66 Instance Total Up Total Up Nbrs ESIs Local Remote
67 __default_evpn__ 0
68 vpls-bgp 2 2 0 0 1 0 0 0
69
70 [edit]
71 regress@rtme-mx-02# run show evpn instance extensive
72 Instance: __default_evpn__
73   Route Distinguisher: 192.168.0.2:0
74   Number of bridge domains: 0
75   Number of neighbors: 0
76
77 Instance: vpls-bgp
78   Route Distinguisher: 192.168.0.2:1
79   VLAN ID: 1234
80   Per-instance MAC route label: 102
81   Duplicate MAC detection threshold: 5
82   Duplicate MAC detection window: 180
83   MAC database status Local Remote
84     MAC advertisements: 0 0
85     MAC+IP advertisements: 0 0
86     Default gateway MAC advertisements: 0 0
87   Number of local interfaces: 2 (2 up)
88     Interface name ESI Mode Status AC-Role
89     .local..8 00:00:00:00:00:00:00:00:00 single-homed Up Root
90     et-0/1/2.1 00:00:00:00:00:00:00:00:00 single-homed Up Root
91   Number of IRB interfaces: 0 (0 up)
92   Number of protect interfaces: 0
93   Number of bridge domains: 1
94     VLAN Domain-ID Intfs/up IRB-intf Mode MAC-sync IM-label MAC-label v4-SG-sync IM-core-NH v6-SG
95     -sync IM-core-NH Trans-ID
96     1234 1 1 Extended Enabled 128 Disabled Disabled
97   Number of neighbors: 1
98     Address MAC MAC+IP AD IM ES Leaf-label Remote-DCI-Peer
99     192.168.0.22 0 0 0 1 0
100   Number of ethernet segments: 0
101   SMET Forwarding: Disabled
102   Migration: BGP-VPLS
103
104 [edit]
105 regress@rtme-mx-02# run show route table bgp.evpn.0
106
107 bgp.evpn.0: 2 destinations, 2 routes (2 active, 0 holddown, 0 hidden)
108 + = Active Route, - = Last Active, * = Both
109
110 3:192.168.0.2:1::1234::192.168.0.2/248 IM
111   * [EVPN/170] 00:21:53
112   Indirect
113 3:192.168.0.22:1::1234::192.168.0.22/248 IM
114   * [BGP/170] 00:07:16, localpref 100, from 192.168.0.22
115     AS path: I, validation-state: unverified
116     > to 11.0.0.2 via et-0/0/3.0, Push 32

```

1.3 LDP-Signaled VPLS to EVPN Migration

All four PEs in the topology in Figure 2 are signalled with LDP-VPLS. We will migrate PEs from VPLS to EVPN one by one. After each migration, we can see how on one end we have VPLS PE and on the other end, we have the EVPN PE.

Here are the configuration changes that need to be done during the migration

1. For the intended routing instance, change the instance-type from “vpls” to “evpn”.
2. Keep the existing VPLS configuration for now.
3. Add protocol evpn and the relevant evpn configurations to the routing instance

4. Add route-distinguisher and route-target to the routing-instance
5. Under the protocol bgp, add the family evpn signalling
6. Commit the configuration

PE1 configuration

```
1 regress@rtme-mx-02# show interfaces
2 et-0/0/2 {
3     description "to rtme-mx-01 et-1/0/1";
4     unit 0 {
5         family inet {
6             address 11.0.0.25/31;
7         }
8         family iso;
9         family mpls
10    }
11 }
12 et-0/0/3 {
13     description "To MX-61 et-0/0/3";
14     unit 0 {
15         family inet {
16             address 11.0.0.3/31;
17         }
18         family iso;
19         family mpls
20    }
21 }
22 et-0/1/2 {
23     description access-qfx16;
24     flexible-vlan-tagging;
25     encapsulation flexible-ethernet-services;
26     unit 1 {
27         encapsulation vlan-vpls;
28         vlan-id 1;
29     }
30 }
31 lo0 {
32     unit 0 {
33         family inet {
34             address 192.168.0.2/32 {
35                 primary;
36             }
37         }
38         family iso;
39     }
40 }
41
42 regress@rtme-mx-02# show routing-options
43 router-id 192.168.0.2;
44 autonomous-system 65000;
45
46 regress@rtme-mx-02# show protocols mpls
47 interface lo0.0;
48 interface all;
49 interface fxp0.0 {
50     disable;
51 }
52
53 regress@rtme-mx-02# show protocols ldp
54 interface all;
55 interface fxp0.0 {
56     disable;
57 }
58 interface lo0.0;
59
60 regress@rtme-mx-02# show protocols bgp
61 group iBGP {
62     type internal;
63     local-address 192.168.0.2;
64     family inet {
65         unicast;
66     }
67     family inet-vpn {
68         unicast;
69     }
70     neighbor 192.168.0.22;
71     neighbor 192.168.0.62;
72     neighbor 192.168.0.61;
```

```

73 }
74
75 regress@rtme-mx-02-1# show routing-options
76 router-id 192.168.0.2;
77 autonomous-system 65000;
78
79 [edit]
80 regress@rtme-mx-02-1# show chassis
81 aggregated-devices {
82     ethernet {
83         device-count 128;
84     }
85 }
86 fpc 0 {
87     pic 0 {
88         pic-mode 100G;
89         tunnel-services;
90     }
91 }
92
93 regress@rtme-mx-02# show routing-instances vpls
94 instance-type vpls;
95 protocols {
96     vpls {
97         neighbor 192.168.0.22;
98         neighbor 192.168.0.61;
99         neighbor 192.168.0.62;
100         vpls-id 2;
101     }
102 }
103 interface et-0/0/2.1;

```

PE3 configuration

```

1 regress@rtme-mx-22# show interfaces
2 et-0/0/2 {
3     description "to rtme-qfx15 et-0/0/2";
4     flexible-vlan-tagging;
5     encapsulation flexible-ethernet-services;
6     unit 1 {
7         encapsulation vlan-vpls;
8         vlan-id 1;
9     }
10 }
11 et-1/0/2 {
12     description "to rtme-mx-62 et-5/2/0";
13     unit 0 {
14         family inet {
15             address 11.0.0.21/31;
16         }
17         family iso;
18         family mpls;
19     }
20 }
21 et-1/0/5 {
22     description "to rtme-mx-01 et-1/0/0";
23     unit 0 {
24         family inet {
25             address 11.0.0.22/31;
26         }
27         family iso;
28         family mpls;
29     }
30 }
31 et-1/1/2 {
32     description "to rtme-mx25 0/0/5";
33     unit 0 {
34         family inet {
35             address 11.0.0.31/31;
36         }
37         family iso;
38         family mpls;
39     }
40 }
41 lo0 {
42     unit 0 {
43         family inet {
44             address 192.168.0.22/32 {
45                 primary;
46             }
47         }
48     }
49 }

```

```

47     }
48     family iso;
49 }
50 }
51
52 regress@rtme-mx-22# show protocols mpls
53 interface all;
54 interface fxp0.0 {
55     disable;
56 }
57
58 regress@rtme-mx-22# show protocols ldp
59 interface all;
60 interface fxp0.0 {
61     disable;
62 }
63
64 regress@rtme-mx-22# show protocols bgp
65 group iBGP {
66     type internal;
67     local-address 192.168.0.22;
68     family inet {
69         unicast;
70     }
71     family inet-vpn {
72         unicast;
73     }
74     neighbor 192.168.0.2;
75     neighbor 192.168.0.61;
76     neighbor 192.168.0.62;
77 }
78
79 regress@rtme-mx-22# show routing-options
80 router-id 192.168.0.22;
81 autonomous-system 65000;
82
83 regress@rtme-mx-22# show chassis
84 aggregated-devices {
85     ethernet {
86         device-count 128;
87     }
88 }
89 fpc 0 {
90     pic 0 {
91         tunnel-services;
92         pic-mode 100G;
93     }
94     pic 1 {
95         tunnel-services;
96         pic-mode 100G;
97     }
98 }
99
100 regress@rtme-mx-22# show routing-instances vpls
101 instance-type vpls;
102 protocols {
103     vpls {
104         neighbor 192.168.0.2;
105         neighbor 192.168.0.61;
106         neighbor 192.168.0.62;
107         vpls-id 22;
108     }
109 }
110 interfaces et-0/0/2.1

```

On PE1, all the VPLS connections are up. Output below shows the status of each PE in the topology. Some output is cut for purpose of brevity.

PE1:

```

1 regress@rtme-mx-02# run show vpls connections
2 Layer-2 VPN connections:
3 Instance: vpls
4   VPLS-id: 2
5   Neighbor Type St Time last up # Up trans
6   192.168.0.22(vpls-id 2) rmt Up Jul 7 00:26:43 2022 1
7   Remote PE: 192.168.0.22, Negotiated control-word: No
8   Incoming label: 47, Outgoing label: 302

```

```

9      Negotiated PW status TLV: No
10     Local interface: vt-0/0/0.1048832, Status: Up, Encapsulation: ETHERNET
11     Description: Intf - vpls vpls neighbor 192.168.0.22 vpls-id 2
12     Flow Label Transmit: No, Flow Label Receive: No
13     192.168.0.61(vpls-id 2) rmt Up Jul 7 00:26:43 2022 1
14     Remote PE: 192.168.0.61, Negotiated control-word: No
15     Incoming label: 48, Outgoing label: 40
16     Negotiated PW status TLV: No
17     Local interface: vt-0/0/0.1048833, Status: Up, Encapsulation: ETHERNET
18     Description: Intf - vpls vpls neighbor 192.168.0.61 vpls-id 2
19     Flow Label Transmit: No, Flow Label Receive: No
20     192.168.0.62(vpls-id 2) rmt Up Jul 7 00:26:43 2022 1
21     Remote PE: 192.168.0.62, Negotiated control-word: No
22     Incoming label: 49, Outgoing label: 16029
23     Negotiated PW status TLV: No
24     Local interface: vt-0/0/0.1048834, Status: Up, Encapsulation: ETHERNET
25     Description: Intf - vpls vpls neighbor 192.168.0.62 vpls-id 2
26     Flow Label Transmit: No, Flow Label Receive: No
27
28 regress@rtme-mx-02# run show vpls mac-table
29 MAC flags (S -static MAC, D -dynamic MAC, L -locally learned, C -Control MAC
30      O -OVSDB MAC, SE -Statistics enabled, NM -Non configured MAC, R -Remote PE MAC, P -Pinned MAC
31      )
32 Routing instance : vpls
33 Bridging domain : __vpls__, VLAN : NA
34 MAC MAC Logical NH MAC active
35 address flags interface Index property source
36 00:cc:cc:cc:00:00 D et-0/1/2.1
37
38 regress@rtme-mx-02-1# run show bgp summary
39 Threading mode: BGP I/O
40 Default eBGP mode: advertise - accept, receive - accept
41 Groups: 1 Peers: 3 Down peers: 0
42 Table Tot Paths Act Paths Suppressed History Damp State Pending
43 inet.0
44      0 0 0 0 0 0
45 bgp.l3vpn.0
46      0 0 0 0 0 0
47 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
48 192.168.0.22 65000 5 3 0 0 26 Establ
49   inet.0: 0/0/0/0
50   bgp.l3vpn.0: 0/0/0/0
51 192.168.0.61 65000 5 3 0 0 26 Establ
52   inet.0: 0/0/0/0
53   bgp.l3vpn.0: 0/0/0/0
54 192.168.0.62 65000 4 3 0 0 26 Establ
55   inet.0: 0/0/0/0
56   bgp.l3vpn.0: 0/0/0/0

```

PE3:

```

1 regress@rtme-mx-22# run show vpls connections
2 Layer-2 VPN connections:
3 Instance: vpls
4 VPLS-id: 2
5 Neighbor Type St Time last up # Up trans
6 192.168.0.2(vpls-id 2) rmt Up Jul 7 00:26:47 2022 1
7 Remote PE: 192.168.0.2, Negotiated control-word: No
8 Incoming label: 302, Outgoing label: 47
9 Negotiated PW status TLV: No
10 Local interface: vt-0/0/0.1049346, Status: Up, Encapsulation: ETHERNET
11 Description: Intf - vpls vpls neighbor 192.168.0.2 vpls-id 2
12 Flow Label Transmit: No, Flow Label Receive: No
13 192.168.0.61(vpls-id 2) rmt Up Jul 7 00:15:19 2022 1
14 Remote PE: 192.168.0.61, Negotiated control-word: No
15 Incoming label: 303, Outgoing label: 41
16 Negotiated PW status TLV: No
17 Local interface: vt-0/0/0.1049345, Status: Up, Encapsulation: ETHERNET
18 Description: Intf - vpls vpls neighbor 192.168.0.61 vpls-id 2
19 Flow Label Transmit: No, Flow Label Receive: No
20 192.168.0.62(vpls-id 2) rmt Up Jul 7 00:07:26 2022 1
21 Remote PE: 192.168.0.62, Negotiated control-word: No
22 Incoming label: 304, Outgoing label: 16030
23 Negotiated PW status TLV: No
24 Local interface: vt-0/0/0.1049344, Status: Up, Encapsulation: ETHERNET
25 Description: Intf - vpls vpls neighbor 192.168.0.62 vpls-id 2
26 Flow Label Transmit: No, Flow Label Receive: No
27
28 regress@rtme-mx-22# run show vpls mac-table

```

```

29 MAC flags (S -static MAC, D -dynamic MAC, L -locally learned, C -Control MAC
30      0 -OVSDB MAC, SE -Statistics enabled, NM -Non configured MAC, R -Remote PE MAC, P -Pinned MAC
31      )
32 Routing instance : vpls
33 Bridging domain : __vpls__, VLAN : NA
34 MAC MAC Logical NH MAC active
35 address flags interface Index property source
36 00:cc:cc:cc:00:00 D vt-0/0/0.1048584
37
38 regress@rtme-mx-22# run show bgp summary
39 Threading mode: BGP I/O
40 Default eBGP mode: advertise - accept, receive - accept
41 Groups: 1 Peers: 3 Down peers: 0
42 Table Tot Paths Act Paths Suppressed History Damp State Pending
43 inet.0
44      0 0 0 0 0 0
45 bgp.l3vpn.0
46      0 0 0 0 0 0
47 lsdist.0
48      0 0 0 0 0 0
49 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
50 192.168.0.2 65000 26 25 0 1 10:27 Establ
51   inet.0: 0/0/0/0
52   bgp.l3vpn.0: 0/0/0/0
53 192.168.0.61 65000 27 25 0 0 10:31 Establ
54   inet.0: 0/0/0/0
55   bgp.l3vpn.0: 0/0/0/0
56 192.168.0.62 65000 27 25 0 0 10:31 Establ
57   inet.0: 0/0/0/0
58   bgp.l3vpn.0: 0/0/0/0

```

On the traffic generator, we verify the packets are flowing end to end:

The screenshot shows a traffic generator interface with a sidebar on the left containing 'Overview', 'Scenario', 'Ports', 'Chassis', 'Protocols', 'Protocol Interfaces', 'BGP/BGP+', 'Static', 'Traffic', 'L2-3 Traffic Items', 'L3VPN_Traffic', and 'L2-3 Flow Groups'. The main area displays 'Traffic Item 1' with a table of flow groups and a detailed statistics table.

Flow Groups	Topology	Endpoint Sets	Tx Port	Rx Ports	Flow Group Name	Configured Frame Size	Applied Frame Size	Frame Rate	Applied Frame Rate	Tx port type
1	✓	✓	rtme-qfx16 xe-0/0/0	rtme-qfx15 xe-0/0/0	Traffic Item 1-EndpointSet...	Fixed: 64	Fixed: 64	1000 fps	1000 fps	Ethernet
2	✓	✓	rtme-qfx15 xe-0/0/0	rtme-qfx16 xe-0/0/0	Traffic Item 1-EndpointSet...	Fixed: 64	Fixed: 64	1000 fps	1000 fps	Ethernet

Traffic Item	Tx Frames	Rx Frames	Frames Del	Loss %	Packet Loss	Tx Frame Rate	Rx Frame Rate	Tx L1 Rate (bps)	Rx L1 Rate (bps)	Rx Bytes	Tx Rate (Bps)	Rx Rate (Bps)	Tx Rate (bps)	Rx Rate (bps)	Tx Rate (Kbps)	Rx Rate (Kbps)
Traffic Item 1	95,762	89,672	6,090	6.360	3,045,000	2,000,000	2,000,000	1,344,000,000	1,344,000,000	5,739,008	128,000,000	128,000,000	1,024,000,000	1,024,000,000	1,024,000	1,024,000

1.3.1 PE1 LDP-VPLS to EVPN Migration

In this section, we are going to migrate the VPLS PE to an EVPN. Only PE1 is migrating to EVPN and the rest will continue to operate as VPLS PEs. After the migration is done the evpn instance shows details about which parameters are programmed including the ESI and the migration model (BGP-VPLS / LDP-VPLS). The highlighted below configurations are the ones that are required for the seamless migration to work. PE1 does move the mac entries from the VPLS database to the EVPN database. However, since there is no other PE in the network that has migrated this PE does not have any EVPN routes learnt.

PE1 configuration

```

1 regress@rtme-mx-02# show | compare
2 [edit routing-instances vpls]
3 - instance-type vpls;
4 + instance-type evpn;
5 [edit routing-instances vpls protocols]
6 + evpn {
7   + interface et-0/1/2.1;
8   + }
9 [edit routing-instances vpls]
10 + route-distinguisher 192.168.0.2:100;
11 + vrf-target target:1:100;
12 [edit protocols bgp group iBGP]
13 + family evpn {
14 +   signaling;
15 + }

```

```

16
17 [edit]
18 regress@rtme-mx-02# show routing-instances vpls
19 instance-type evpn;
20 protocols {
21     evpn {
22         interface et-0/1/2.1;
23     }
24     vpls {
25         neighbor 192.168.0.22;
26         neighbor 192.168.0.61;
27         neighbor 192.168.0.62;
28         vpls-id 2;
29     }
30 }
31 interface et-0/1/2.1;
32 route-distinguisher 192.168.0.2:100;
33 vrf-target target:1:100;
34
35 regress@rtme-mx-02# show protocols bgp
36 group iBGP {
37     type internal;
38     local-address 192.168.0.2;
39     family inet {
40         unicast;
41     }
42     family inet-vpn {
43         unicast;
44     }
45     family evpn {
46         signaling;
47     }
48     neighbor 192.168.0.22;
49     neighbor 192.168.0.62;
50     neighbor 192.168.0.61;
51 }

```

PE1 operational status

```

1 regress@rtme-mx-02# run show evpn instance vpls extensive
2 Instance: vpls
3 Route Distinguisher: 192.168.0.2:100
4 VLAN ID: 1
5 Per-instance MAC route label: 101
6 Duplicate MAC detection threshold: 5
7 Duplicate MAC detection window: 180
8 MAC database status Local Remote
9 MAC advertisements: 1 0
10 MAC+IP advertisements: 0 0
11 Default gateway MAC advertisements: 0 0
12 Number of local interfaces: 2 (2 up)
13 Interface name ESI Mode Status AC-Role
14 .local..14 00:00:00:00:00:00:00:00:00 single-homed Up Root
15 et-0/1/2.1 00:00:00:00:00:00:00:00:00 single-homed Up Root
16 Number of IRB interfaces: 0 (0 up)
17 Number of protect interfaces: 0
18 Number of bridge domains: 1
19 VLAN Domain-ID Intfs/up IRB-intf Mode MAC-sync v4-SG-sync v6-SG-sync
20 1 1 1 Extended Enabled Disabled Disabled
21 Number of neighbors: 0
22 Number of ethernet segments: 0
23 SMET Forwarding: Disabled
24 Migration: LDP-VPLS
25
26 regress@rtme-mx-02# run show vpls connections
27 Instance: vpls
28 VPLS-id: 2
29 Neighbor Type St Time last up # Up trans
30 192.168.0.22(vpls-id 2) rmt Up Jul 12 02:10:24 2022 1
31 Remote PE: 192.168.0.22, Negotiated control-word: No
32 Incoming label: 103, Outgoing label: 28
33 Negotiated PW status TLV: No
34 Local interface: vt-0/0/0.1049106, Status: Up, Encapsulation: ETHERNET
35 Description: Intf - vpls vpls neighbor 192.168.0.22 vpls-id 2
36 Flow Label Transmit: No, Flow Label Receive: No
37 192.168.0.61(vpls-id 2) rmt Up Jul 12 02:10:24 2022 1
38 Remote PE: 192.168.0.61, Negotiated control-word: No
39 Incoming label: 104, Outgoing label: 42
40 Negotiated PW status TLV: No
41 Local interface: vt-0/0/0.1049105, Status: Up, Encapsulation: ETHERNET

```

```

42      Description: Intf - vpls vpls neighbor 192.168.0.61 vpls-id 2
43      Flow Label Transmit: No, Flow Label Receive: No
44      192.168.0.62(vpls-id 2) rmt Up Jul 12 02:10:24 2022 1
45      Remote PE: 192.168.0.62, Negotiated control-word: No
46      Incoming label: 105, Outgoing label: 16029
47      Negotiated PW status TLV: No
48      Local interface: vt-0/0/0.1049104, Status: Up, Encapsulation: ETHERNET
49      Description: Intf - vpls vpls neighbor 192.168.0.62 vpls-id 2
50      Flow Label Transmit: No, Flow Label Receive: No
51
52 regress@rtme-mx-02# run show vpls mac-table
53
54
55 regress@rtme-mx-02# run show evpn mac-table
56 MAC flags (S -static MAC, D -dynamic MAC, L -locally learned, C -Control MAC
57      O -OVSDB MAC, SE -Statistics enabled, NM -Non configured MAC, R -Remote PE MAC, P -Pinned MAC
58      )
59 Routing instance : vpls
60 Bridging domain : __vpls__, VLAN : 1
61 MAC MAC Logical NH MAC active
62 address flags interface Index property source
63 00:cc:cc:cc:00:00 D et-0/1/2.1
64
65 regress@rtme-mx-02-1# run show evpn database
66 Instance: vpls
67 VLAN DomainId MAC address Active source Timestamp IP address
68 1 00:cc:cc:cc:00:00 et-0/1/2.1 Jul 12 02:10:24
69
70 regress@rtme-mx-02-1# run show bgp summary
71 Threading mode: BGP I/O
72 Default eBGP mode: advertise - accept, receive - accept
73 Groups: 1 Peers: 3 Down peers: 0
74 Table Tot Paths Act Paths Suppressed History Damp State Pending
75 inet.0
76      0 0 0 0 0 0
77
78 bgp.l3vpn.0
79      0 0 0 0 0 0
80
81 bgp.evpn.0
82      0 0 0 0 0 0
83
84 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
85 192.168.0.22 65000 30 29 0 0 12:11 Establ
86      inet.0: 0/0/0/0
87      bgp.l3vpn.0: 0/0/0/0
88 192.168.0.61 65000 31 29 0 0 12:11 Establ
89      inet.0: 0/0/0/0
90      bgp.l3vpn.0: 0/0/0/0
91 192.168.0.62 65000 31 29 0 0 12:11 Establ
92      inet.0: 0/0/0/0
93      bgp.l3vpn.0: 0/0/0/0

```

PE3 operational status

```

1 regress@rtme-mx-22> show vpls connections
2 Instance: vpls
3 VPLS-id: 2
4 Neighbor Type St Time last up # Up trans
5 192.168.0.2(vpls-id 2) rmt Up Jul 12 01:48:49 2022 1
6 Remote PE: 192.168.0.2, Negotiated control-word: No
7 Incoming label: 28, Outgoing label: 103
8 Negotiated PW status TLV: No
9 Local interface: vt-0/0/0.1048584, Status: Up, Encapsulation: ETHERNET
10 Description: Intf - vpls vpls neighbor 192.168.0.2 vpls-id 2
11 Flow Label Transmit: No, Flow Label Receive: No
12 192.168.0.61(vpls-id 2) rmt Up Jul 12 01:48:49 2022 1
13 Remote PE: 192.168.0.61, Negotiated control-word: No
14 Incoming label: 29, Outgoing label: 41
15 Negotiated PW status TLV: No
16 Local interface: vt-0/0/0.1048586, Status: Up, Encapsulation: ETHERNET
17 Description: Intf - vpls vpls neighbor 192.168.0.61 vpls-id 2
18 Flow Label Transmit: No, Flow Label Receive: No
19 192.168.0.62(vpls-id 2) rmt Up Jul 12 01:48:49 2022 1
20 Remote PE: 192.168.0.62, Negotiated control-word: No
21 Incoming label: 30, Outgoing label: 16030
22 Negotiated PW status TLV: No
23 Local interface: vt-0/0/0.1048585, Status: Up, Encapsulation: ETHERNET
24 Description: Intf - vpls vpls neighbor 192.168.0.62 vpls-id 2
25 Flow Label Transmit: No, Flow Label Receive: No
26
27 regress@rtme-mx-22> show vpls mac-table
28 MAC flags (S -static MAC, D -dynamic MAC, L -locally learned, C -Control MAC

```

```

29 0 -OVSDB MAC, SE -Statistics enabled, NM -Non configured MAC, R -Remote PE MAC, P -Pinned MAC
30 )
31 Routing instance : vpls
32 Bridging domain : __vpls__, VLAN : NA
33 MAC MAC Logical NH MAC active
34 address flags interface Index property source
35 00:cc:cc:cc:00:00 D vt-0/0/0.1048584

```

Screenshot of the traffic generator console while PE1 migrates showing a slight disruption because of BGP signalling.

The screenshot shows a traffic generator interface with a sidebar on the left containing 'Overview', 'Scenario', 'Ports', 'Chassis', 'Protocols', 'BGP/BGP+', 'Static', and 'Traffic'. The main area displays 'Flow Groups' and 'Traffic Item Statistics'.

Flow Groups	Topology	Endpoint Sets	Transmit State	Suspend	Tx Port	Rx Ports	Flow Group Name	Configured Frame Size	Applied Frame Size	Frame Rate	Applied Frame Rate	Tx port type
1	✓	✓	✓	✓	rtme-qfx 15 xe-0/0/3:0	rtme-qfx 15 xe-0/0/3:0	Traffic Item 1-EndpointSet...	Fixed: 64	Fixed: 64	1000 fps	1000 fps	Ethernet
2	✓	✓	✓	✓	rtme-qfx 16 xe-0/0/0:0	rtme-qfx 16 xe-0/0/0:0	Traffic Item 1-EndpointSet...	Fixed: 64	Fixed: 64	1000 fps	1000 fps	Ethernet

Traffic Item	Tx Frames	Rx Frames	Frames Del	Loss %	Packet Loss	Tx Frame Rate	Rx Frame Rate	Tx L1 Rate (bps)	Rx L1 Rate (bps)	Rx Bytes	Tx Rate (Bps)	Rx Rate (Bps)	Tx Rate (bps)	Rx Rate (bps)	Tx Rate (Kbps)	Rx Rate (Kbps)
Traffic Item 1	972,288	964,852	7,436	0.765	3,718.000	2,000.000	2,000.000	1,344,000.000	1,344,000.000	61,750,528	128,000.000	128,000.000	1,024,000.000	1,024,000.000	1,024.000	1,024.000

1.3.2 PE3 LDP-VPLS to EVPN Migration

In this section, we are going to migrate the VPLS PE (PE3) to an EVPN PE. As seen in the section above PE1 has already migrated to EVPN, so when PE3 migrates the rest will continue to operate as VPLS PEs. After completion of PE1 & PE3 migration, these will continue to be EVPN PE and the other two PEs (PE2 & PE4) will continue to operate as VPLS PE.

PE3 configuration

```

1 regress@rtme-mx-22# show | compare
2 [edit routing-instances vpls]
3 - instance-type vpls;
4 + instance-type evpn;
5 [edit routing-instances vpls protocols]
6 + evpn {
7 + interface et-0/0/2.1;
8 + }
9 [edit routing-instances vpls]
10 + route-distinguisher 192.168.0.22:100;
11 + vrf-target target:1:100;
12 [edit protocols bgp group iBGP]
13 + family evpn {
14 + signaling;
15 + }
16
17 regress@rtme-mx-22# show routing-instances vpls
18 instance-type evpn;
19 protocols {
20   evpn {
21     interface et-0/0/2.1;
22   }
23   vpls {
24     neighbor 192.168.0.2;
25     neighbor 192.168.0.61;
26     neighbor 192.168.0.62;
27     vpls-id 2;
28   }
29 }
30 interface et-0/0/2.1;
31 route-distinguisher 192.168.0.22:100;
32 vrf-target target:1:100;
33
34 regress@rtme-mx-22# commit
35 commit complete

```

PE1 operational status

```

1 regress@rtme-mx-02# run show bgp summary
2 Threading mode: BGP I/O
3 Default eBGP mode: advertise - accept, receive - accept
4 Groups: 1 Peers: 3 Down peers: 0
5 Table Tot Paths Act Paths Suppressed History Damp State Pending
6 inet.0
7         0 0 0 0 0 0
8 bgp.l3vpn.0
9         0 0 0 0 0 0
10 bgp.evpn.0
11        1 1 0 0 0 0
12 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
13 192.168.0.22 65000 11 11 0 1 2:23 Establ
14   inet.0: 0/0/0/0
15   bgp.l3vpn.0: 0/0/0/0
16   bgp.evpn.0: 1/1/1/0
17   vpls.evpn.0: 1/1/1/0
18   __default_evpn__.evpn.0: 0/0/0/0
19 192.168.0.61 65000 60 58 0 0 25:21 Establ
20   inet.0: 0/0/0/0
21   bgp.l3vpn.0: 0/0/0/0
22 192.168.0.62 65000 60 59 0 0 25:21 Establ
23   inet.0: 0/0/0/0
24   bgp.l3vpn.0: 0/0/0/0
25
26 regress@rtme-mx-02# run show evpn instance vpls extensive
27 Instance: vpls
28   Route Distinguisher: 192.168.0.2:100
29   VLAN ID: 1
30   Per-instance MAC route label: 101
31   Duplicate MAC detection threshold: 5
32   Duplicate MAC detection window: 180
33   MAC database status Local Remote
34   MAC advertisements: 1 0
35   MAC+IP advertisements: 0 0
36   Default gateway MAC advertisements: 0 0
37   Number of local interfaces: 2 (2 up)
38   Interface name ESI Mode Status AC-Role
39   .local..14 00:00:00:00:00:00:00:00:00:00:00 single-homed Up Root
40   et-0/1/2.1 00:00:00:00:00:00:00:00:00:00:00 single-homed Up Root
41   Number of IRB interfaces: 0 (0 up)
42   Number of protect interfaces: 0
43   Number of bridge domains: 1
44   VLAN Domain-ID Intfs/up IRB-intf Mode MAC-sync v4-SG-sync v6-SG-sync
45   1 1 1 Extended Enabled Disabled Disabled
46   Number of neighbors: 1
47   Address MAC MAC+IP AD IM ES Leaf-label Remote-DCI-Peer
48   192.168.0.22 0 0 0 1 0
49   Number of ethernet segments: 0
50   SMET Forwarding: Disabled
51   Migration: LDP-VPLS
52
53 regress@rtme-mx-02# run show evpn database
54 Instance: vpls
55 VLAN DomainId MAC address Active source Timestamp IP address
56 1 00:cc:cc:cc:00:00 et-0/1/2.1 Jul 12 02:33:27
57
58 regress@rtme-mx-02-1# run show evpn mac-table
59 MAC flags (S -static MAC, D -dynamic MAC, L -locally learned, C -Control MAC
60   0 -OVSDB MAC, SE -Statistics enabled, NM -Non configured MAC, R -Remote PE MAC, P -Pinned MAC
61   )
62 Routing instance : vpls
63 Bridging domain : __vpls__, VLAN : 1
64   MAC MAC Logical NH MAC active
65   address flags interface Index property source
66   00:cc:cc:cc:00:00 D et-0/1/2.1
67
68 regress@rtme-mx-02# run show route table vpls.evpn.0
69 vpls.evpn.0: 3 destinations, 3 routes (3 active, 0 holddown, 0 hidden)
70 + = Active Route, - = Last Active, * = Both
71
72 2:192.168.0.2:100::1::00:cc:cc:cc:00:00/304 MAC/IP
73   *[EVPN/170] 00:26:03
74   Indirect
75 3:192.168.0.2:100::1::192.168.0.2/248 IM
76   *[EVPN/170] 00:26:01
77   Indirect
78 3:192.168.0.22:100::1::192.168.0.22/248 IM
79   *[BGP/170] 00:03:03, localpref 100, from 192.168.0.22
80   AS path: I, validation-state: unverified

```

```

81 > to 11.0.0.24 via et-0/0/2.0, Push 313
82
83 regress@rtme-mx-02-1# run show route table bgp.evpn.0
84 bgp.evpn.0: 3 destinations, 3 routes (3 active, 0 holddown, 0 hidden)
85 + = Active Route, - = Last Active, * = Both
86 2:192.168.0.2:100::1::00:cc:cc:cc:00:00/304 MAC/IP
87      *[EVPN/170] 00:26:08
88      Indirect
89 3:192.168.0.2:100::1::192.168.0.2/248 IM
90      *[EVPN/170] 00:26:06
91      Indirect
92 3:192.168.0.22:100::1::192.168.0.22/248 IM
93      *[BGP/170] 00:03:08, localpref 100, from 192.168.0.22
94      AS path: I, validation-state: unverified
95 > to 11.0.0.24 via et-0/0/2.0, Push 313

```

PE3 operational status:

```

1 regress@rtme-mx-22# run show bgp summary
2 Threading mode: BGP I/O
3 Default eBGP mode: advertise - accept, receive - accept
4 Groups: 1 Peers: 3 Down peers: 0
5 Table Tot Paths Act Paths Suppressed History Damp State Pending
6 inet.0
7      0 0 0 0 0 0
8 bgp.l3vpn.0
9      0 0 0 0 0 0
10 lsdist.0
11      0 0 0 0 0 0
12 bgp.evpn.0
13      2 2 0 0 0 0
14 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
15 192.168.0.2 65000 7 5 0 0 3 Establ
16   inet.0: 0/0/0/0
17   bgp.l3vpn.0: 0/0/0/0
18   bgp.evpn.0: 2/2/2/0
19   vpls.evpn.0: 2/2/2/0
20   __default_evpn__.evpn.0: 0/0/0/0
21 192.168.0.61 65000 4 3 0 0 3 Establ
22   inet.0: 0/0/0/0
23   bgp.l3vpn.0: 0/0/0/0
24 192.168.0.62 65000 4 3 0 0 3 Establ
25   inet.0: 0/0/0/0
26   bgp.l3vpn.0: 0/0/0/0
27
28 regress@rtme-mx-22# run show evpn instance vpls extensive
29 Instance: vpls
30 Route Distinguisher: 192.168.0.22:100
31 VLAN ID: 1
32 Per-instance MAC route label: 36
33 Duplicate MAC detection threshold: 5
34 Duplicate MAC detection window: 180
35 MAC database status Local Remote
36 MAC advertisements: 0 1
37 MAC+IP advertisements: 0 0
38 Default gateway MAC advertisements: 0 0
39 Number of local interfaces: 2 (2 up)
40 Interface name ESI Mode Status AC-Role
41 .local..9 00:00:00:00:00:00:00:00:00:00 single-homed Up Root
42 et-0/0/2.1 00:00:00:00:00:00:00:00:00:00 single-homed Up Root
43 Number of IRB interfaces: 0 (0 up)
44 Number of protect interfaces: 0
45 Number of bridge domains: 1
46   VLAN Domain-ID Intfs/up IRB-intf Mode MAC-sync v4-SG-sync v6-SG-sync
47   1 1 1 Extended Enabled Disabled Disabled
48 Number of neighbors: 1
49   Address MAC MAC+IP AD IM ES Leaf-label Remote-DCI-Peer
50   192.168.0.2 1 0 0 1 0
51 Number of ethernet segments: 0
52 SMET Forwarding: Disabled
53 Migration: LDP-VPLS
54
55 regress@rtme-mx-22# run show evpn database
56 Instance: vpls
57 VLAN DomainId MAC address Active source Timestamp IP address
58 1 00:cc:cc:cc:00:00 192.168.0.2 Jul 12 02:33:25
59
60 regress@rtme-mx-22# run show evpn mac-table
61 MAC flags (S -static MAC, D -dynamic MAC, L -locally learned, C -Control MAC
62   O -OVSDB MAC, SE -Statistics enabled, NM -Non configured MAC, R -Remote PE MAC, P -Pinned MAC

```

```

63      )
64  Routing instance : vpls
65  Bridging domain : __vpls__, VLAN : 1
66  MAC MAC Logical NH MAC active
67  address flags interface Index property source
68  00:cc:cc:cc:00:00 DC 1048578 192.168.0.2
69
70 regress@rtme-mx-22# run show route table vpls.evpn.0
71 vpls.evpn.0: 3 destinations, 3 routes (3 active, 0 holddown, 0 hidden)
72 + = Active Route, - = Last Active, * = Both
73
74 2:192.168.0.2:100::1::00:cc:cc:cc:00:00/304 MAC/IP
75   * [BGP/170] 00:01:32, localpref 100, from 192.168.0.2
76   AS path: I, validation-state: unverified
77   > to 11.0.0.23 via et-1/0/5.0, Push 305
78 3:192.168.0.2:100::1::192.168.0.2/248 IM
79   * [BGP/170] 00:01:32, localpref 100, from 192.168.0.2
80   AS path: I, validation-state: unverified
81   > to 11.0.0.23 via et-1/0/5.0, Push 305
82 3:192.168.0.22:100::1::192.168.0.22/248 IM
83   * [EVPN/170] 00:01:33
84   Indirect
85
86 regress@rtme-mx-22# run show route table bgp.evpn.0
87 bgp.evpn.0: 3 destinations, 3 routes (3 active, 0 holddown, 0 hidden)
88 + = Active Route, - = Last Active, * = Both
89
90 2:192.168.0.2:100::1::00:cc:cc:cc:00:00/304 MAC/IP
91   * [BGP/170] 00:01:36, localpref 100, from 192.168.0.2
92   AS path: I, validation-state: unverified
93   > to 11.0.0.23 via et-1/0/5.0, Push 305
94 3:192.168.0.2:100::1::192.168.0.2/248 IM
95   * [BGP/170] 00:01:36, localpref 100, from 192.168.0.2
96   AS path: I, validation-state: unverified
97   > to 11.0.0.23 via et-1/0/5.0, Push 305
98 3:192.168.0.22:100::1::192.168.0.22/248 IM
99   * [EVPN/170] 00:01:37
100  Indirect

```

Screenshot of traffic generator while PE1 migrates showing a slight disruption because of BGP signalling.

Flow Groups	Enabled	Topology	Endpoint Sets	Tx Port	Rx Ports	Flow Group Name	Configured Frame Size	Applied Frame Size	Frame Rate	Applied Frame Rate	Tx port type
1	✓	✓	✓	rtme-qfx16 xe-0/0/0	rtme-qfx15 xe-0/0/0	Traffic Item 1-EndpointSet...	Fixed: 64	Fixed: 64	1000 fps	1000 fps	Ethernet
2	✓	✓	✓	rtme-qfx15 xe-0/0/0	rtme-qfx16 xe-0/0/0	Traffic Item 1-EndpointSet...	Fixed: 64	Fixed: 64	1000 fps	1000 fps	Ethernet

Traffic Item	Tx Frames	Rx Frames	Frames Del	Loss %	Packet Loss	Tx Frame Rate	Rx Frame Rate	Tx L1 Rate (bps)	Rx L1 Rate (bps)	Rx Bytes	Tx Rate (Bps)	Rx Rate (Bps)	Tx Rate (bps)	Rx Rate (bps)	Tx Rate (Kbps)	Rx Rate (Kbps)
Traffic Item 1	95,762	89,672	6,090	6.360	3,045,000	2,000,000	2,000,000	1,344,000,000	1,344,000,000	5,739,008	128,000,000	128,000,000	1,024,000,000	1,024,000,000	1,024,000	1,024,000

1.3.3 PE1 and PE3 Configuration Clean Up

Once the migration of the PEs is complete, we will remove the VPLS related configuration from these PEs.

PE1 configuration

```

1 [edit]
2 regress@rtme-mx-02# show routing-instances vpls
3 instance-type evpn;
4 protocols {
5   evpn {
6     interface et-0/1/2.1;
7   }
8   vpls {
9     neighbor 192.168.0.22;
10    neighbor 192.168.0.61;
11    neighbor 192.168.0.62;
12    vpls-id 2;
13  }
14 }
15 interface et-0/1/2.1;
16 route-distinguisher 192.168.0.2:100;

```

```

17 vrf-target target:1:100;
18
19 regress@rtme-mx-02-1# delete routing-instances vpls protocols vpls
20
21 [edit]
22 regress@rtme-mx-02-1# show | compare
23 [edit routing-instances vpls protocols]
24 - vpls {
25 - neighbor 192.168.0.22;
26 - neighbor 192.168.0.61;
27 - neighbor 192.168.0.62;
28 - vpls-id 2;
29 - }
30
31 [edit]
32 regress@rtme-mx-02-1# commit
33 commit complete
34
35 [edit]
36 regress@rtme-mx-02-1# show routing-instances vpls
37 instance-type evpn;
38 protocols {
39     evpn {
40         interface et-0/1/2.1;
41     }
42 }
43 interface et-0/1/2.1;
44 route-distinguisher 192.168.0.2:100;
45 vrf-target target:1:100;

```

Screenshot of traffic generator: traffic disruption when reverting to VPLS from EVPN on PE1

Enabled	Transmit State	Suspend	Tx Port	Rx Ports	Flow Group Name	Configured Frame Size	Applied Frame Size	Frame Rate	Applied Frame Rate	Tx port type
1	✓	✗	rtme-qfx16 xe-0/0/0	rtme-qfx15 xe-0/0/3/0	Traffic Item 1-EndpointSet...	Fixed: 64	Fixed: 64	1000 fps	1000 fps	Ethernet
2	✓	✗	rtme-qfx15 xe-0/0/3/0	rtme-qfx16 xe-0/0/0	Traffic Item 1-EndpointSet...	Fixed: 64	Fixed: 64	1000 fps	1000 fps	Ethernet

Traffic Item	Tx Frames	Rx Frames	Frames Delta	Loss %	Packet Loss Duration (ms)	Tx Frame Rate	Rx Frame Rate	Tx L1 Rate (Bps)	Rx L1 Rate (Bps)	Rx Bytes	Tx Rate (Bps)	Rx Rate (Bps)	Tx Rate (Kbps)	Rx Rate (Kbps)	Tx Rate (Mbps)
Traffic Item 1	303,940	293,966	9,974	3.282	4,987.000	2,000.000	2,000.000	1,344,000.000	1,344,000.000	18,813,824	128,000.000	128,000.000	1,024,000.000	1,024,000.000	1,024.000

And reverting to VPLS from EVPN on PE3

Enabled	Transmit State	Suspend	Tx Port	Rx Ports	Flow Group Name	Configured Frame Size	Applied Frame Size	Frame Rate	Applied Frame Rate	Tx port type
1	✓	✗	rtme-qfx16 xe-0/0/0	rtme-qfx15 xe-0/0/3/0	Traffic Item 1-EndpointSet...	Fixed: 64	Fixed: 64	1000 fps	1000 fps	Ethernet
2	✓	✗	rtme-qfx15 xe-0/0/3/0	rtme-qfx16 xe-0/0/0	Traffic Item 1-EndpointSet...	Fixed: 64	Fixed: 64	1000 fps	1000 fps	Ethernet

Traffic Item	Tx Frames	Rx Frames	Frames Delta	Loss %	Packet Loss Duration (ms)	Tx Frame Rate	Rx Frame Rate	Tx L1 Rate (Bps)	Rx L1 Rate (Bps)	Rx Bytes	Tx Rate (Bps)	Rx Rate (Bps)	Tx Rate (Kbps)	Rx Rate (Kbps)	Tx Rate (Mbps)
Traffic Item 1	223,892	214,375	9,517	4.251	4,758.500	2,000.000	2,000.000	1,344,000.000	1,344,000.000	13,720,000	128,000.000	128,000.000	1,024,000.000	1,024,000.000	1,024.000

1.4 References

- TechLibrary: <https://www.juniper.net/documentation/us/en/software/junos/evpn-vxlan/topics/concept/bgp-vpls-to-evpn-migration.html>

1.5 Glossary

- CE: Customer Edge
- EVPN: Ethernet Virtual Private Network
- MAN: Metro Area Network
- MBH: Mobile BackHaul Network

- PE: Provider Edge
- TGEN: Traffic Generator
- VPLS: Virtual Private LAN Services

1.6 Acknowledgement

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